MECHANICS APPLIED TO AEROSPACE ENGINEERING - PART I

# Topic 4: Angular Momentum and Central Forces - Problem Sets

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Source: sources/cuatrimestre-1/03-engineering-mechanics/problemas/Problems.pdf (2026 Edition)

**Statements and solutions:** Each card gives the statement from the official collection (with the PDF page) and a worked solution in four phases: hypotheses and data, frames and change-of-basis matrices, step-by-step derivation, and interpretation with unit and limit checks. Only Problems 17 and 39 (part f) have an official key entry; the other results are derived here and checked numerically.

PROBLEM 42

## Particle Attracted to the Origin by $F = m\mu/r^{\alpha}$

Original statement · Problems.pdf, p. 29

Binet + Effective Potential

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*(Advanced problem.)* A point particle $P$ of mass $m$ is attracted by the origin of coordinates $O$ of the inertial reference frame $S_0$ by a force with magnitude

$$F = \frac{m\mu}{r^{\alpha}}$$

where $r$ is the distance between $O$ and $P$, and $\mu, \alpha$ are known positive constants.

**Notation.**

The official text writes $F = m\alpha/r^{\beta}$ (strength $\alpha$, exponent $\beta$) and asks for $\beta = 1, 2, 3, 4$. Here the strength is called $\mu$ and the exponent $\alpha$, so the cases "$\alpha = 1, 2, 3, 4$" below are the official $\beta = 1, 2, 3, 4$, and $\mu/h^2$ is the official $\alpha/h^2$.

**(a)** Show that the angular momentum vector and the mechanical energy of $P$ are conserved quantities of motion, and that the motion of $P$ occurs in a plane.

**(b)** Using polar coordinates $(r, \theta)$ in the plane of motion, find a second-order differential equation to solve for $u \equiv 1/r$ as a function of $\theta$ (i.e., $u = u(\theta)$).

**(c)** Find a first integral of this equation and discuss the limits of motion of $u$ as a function of $\mu/h^2$, $\alpha$, and the values of $u$ and $u' \equiv du/d\theta$ at $t = 0$, where $h$ is the magnitude of the angular momentum vector about the origin per unit mass. For $\alpha = 1, 2, 3$, and $4$, determine under which conditions: (i) the particle escapes to infinity ($u \to 0$); (ii) the particle reaches $r = 0$ ($u \to \infty$); (iii) the particle oscillates between two values of $u$.

*Hint: try to solve first the problem with $\alpha = 2$, and then try the other values.*

**Worked Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **System:** a particle $P$ of mass $m$ attracted to $O$ by $\mathbf{F} = -\dfrac{m\mu}{r^{\alpha}}\mathbf{e}_r$.
- **Hypotheses:** $S_0$ inertial; $\mu > 0, \alpha > 0$ constants; no other forces; $h \neq 0$ unless stated.
- **Degrees of freedom:** free particle, $\mathrm{CDOF} = 3$; after part (a) the motion is planar, $\mathrm{CDOF} = 2$ with coordinates $(r, \theta)$.
- **Variables:** $u(\theta) = 1/r(\theta), u' = du/d\theta, h = \|\mathbf{r} \times \mathbf{v}\|$ [$\mathrm{m}^2/\mathrm{s}$].

| Symbol | Meaning | SI unit |
|---|---|---|
| $\mu$ | force strength per unit mass, $F/m = \mu/r^{\alpha}$ | $\mathrm{m}^{\alpha+1}/\mathrm{s}^2$ |

| Symbol | Meaning | SI unit |
|---|---|---|
| $h$ | specific angular momentum | $\mathrm{m^2/s}$ |
| $u$ | reciprocal radius | $\mathrm{m^{-1}}$ |
| $\xi$ | specific mechanical energy | J/kg = $\mathrm{m^2/s^2}$ |

PHASE 2

### Reference Frames, Vector Bases & Coordinate Geometry

- **Change of basis (polar basis in the plane of motion).** Take $\mathbf{k}$ along $\mathbf{H}_O$. The polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\mathcal{B}_0 = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

The angular velocity of $\mathcal{B}_1$ is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$; Poisson's formula $\dot{\mathbf{e}} = \boldsymbol{\omega} \times \mathbf{e}$ gives $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$. The product rule on $\mathbf{r} = r\mathbf{e}_r$ yields

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta, \qquad \mathbf{a} = \left(\ddot r - r\dot\theta^2\right)\mathbf{e}_r + \left(r\ddot\theta + 2\dot r\dot\theta\right)\mathbf{e}_\theta$$

PHASE 3

### Step-by-Step Mathematical Deduction

#### Part (a): conservation laws and planar motion

**Pedagogical Justification:** The force line of action passes through the fixed point $O$, so its torque about $O$ vanishes, and the angular momentum theorem about a fixed point (Notes Eq. 4.53) gives a constant $\mathbf{H}_O$.

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = r\mathbf{e}_r \times \left(-\frac{m\mu}{r^\alpha}\mathbf{e}_r\right) = \mathbf{0} \implies \left.\frac{d\mathbf{H}_O}{dt}\right|_0 = \mathbf{0} \implies \mathbf{H}_O = m\,\mathbf{r} \times \mathbf{v} = \text{const}$$

Since $\mathbf{r}\cdot\mathbf{H}_O \equiv 0$ and $\mathbf{v}\cdot\mathbf{H}_O \equiv 0$, $P$ stays in the plane through $O$ perpendicular to $\mathbf{H}_O$.

**Pedagogical Justification:** The force is purely radial and depends only on $r$, so its elementary work $\mathbf{F}\cdot d\mathbf{r} = F(r)\,dr$ is the exact differential of a function of $r$ (equivalently $\nabla\times\mathbf{F}=\mathbf{0}$). The force is conservative, so the energy is conserved. The potential energy per unit mass follows from Barrow's rule with the reference at infinity (for $\alpha>1$; for $\alpha\le 1$ the reference point is arbitrary):

$$V_0(r) = -\int_{\infty}^{r}\left(-\frac{\mu}{s^{\alpha}}\right)ds = \mu\left[\frac{s^{1-\alpha}}{1-\alpha}\right]_{\infty}^{r} = -\frac{\mu}{(\alpha-1)\,r^{\alpha-1}} \quad (\alpha>1), \qquad V_0(r)=\mu\ln r \quad (\alpha=1)$$

$$\xi = \tfrac{1}{2}v^2 + V_0(r) = \text{const}$$

**Part (b): Binet equation for a power law**

**Pedagogical Justification:** Eliminate time with $h = r^2\dot\theta = \text{const}$, i.e. $\dot\theta = hu^2$. The chain rule for $r = u^{-1}$ gives $dr/d\theta = -u'/u^2$.

$$\dot r = \frac{dr}{d\theta}\dot\theta = \left(-\frac{u'}{u^2}\right)\left(hu^2\right) = -hu', \qquad \ddot r = \frac{d}{dt}\left(-hu'\right) = -hu''\,\dot\theta = -h^2u^2u''$$

The radial equation $\ddot r - r\dot\theta^2 = -\mu/r^{\alpha} = -\mu u^{\alpha}$ with $r\dot\theta^2 = \frac{1}{u}(hu^2)^2 = h^2u^3$ becomes

$$-h^2u^2u'' - h^2u^3 = -\mu u^{\alpha} \quad \overset{\div(-h^2u^2)}{\Longrightarrow} \quad \boxed{u'' + u = \frac{\mu}{h^2}\,u^{\alpha-2}}$$

**Part (c): first integral**

> **Pedagogical Justification:** Write the specific energy with $\dot{r} = -hu'$ and $r\dot{\theta} = hu$. Since $\xi$ is constant, this is a first integral; it can be checked by multiplying the Binet equation by $u'$ and integrating in $\theta$.

$$\xi = \tfrac{1}{2}h^2\left[(u')^2 + u^2\right] + V_0\left(\frac{1}{u}\right) \implies \boxed{(u')^2 + W_{\text{eff}}(u) = E^*}, \quad E^* \equiv \frac{2\xi}{h^2}, \quad W_{\text{eff}}(u) = u^2 + \frac{2V_0(1/u)}{h^2}$$

Check: $\frac{d}{d\theta}\left[(u')^2 + W_{\text{eff}}\right] = 2u'u'' + W'_{\text{eff}}u' = 0$ gives $u'' = -\frac{1}{2}W'_{\text{eff}}$, which is the Binet equation. Motion exists only where $W_{\text{eff}}(u) \le E^*$; turning points ($u' = 0$) are the roots of $W_{\text{eff}}(u) = E^*$, and a minimum or a maximum of $W_{\text{eff}}$ is a circular orbit.

**Case $\alpha = 2$: inverse square (Kepler)**

- Binet: $u'' + u = \mu/h^2$. Potential: $V_0 = -\mu u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2}u = \left(u - \dfrac{\mu}{h^2}\right)^2 - \left(\dfrac{\mu}{h^2}\right)^2$, a parabola with minimum $-(\mu/h^2)^2$ at $u = \mu/h^2$.
- **(i) Escape ($u \to 0$):** possible if $E^* \ge W_{\text{eff}}(0) = 0$, i.e. $\xi \ge 0$ (parabola or hyperbola).
- **(ii) Reaching $r = 0$:** impossible for $h \neq 0$, because $W_{\text{eff}} \to +\infty$ as $u \to \infty$.
- **(iii) Oscillation:** for $-(\mu/h^2)^2 < E^* < 0$, i.e. $-\mu^2/(2h^2) < \xi < 0$: $u$ oscillates between the two roots (periapsis and apoapsis of an ellipse); $E^* = -(\mu/h^2)^2$ is the circular orbit.

**Case $\alpha = 1$: $F \propto 1/r$**

- Binet: $u'' + u = \dfrac{\mu}{h^2 u}$. Potential $V_0(1/u) = -\mu \ln u$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{h^2}\ln u$.
- $W_{\text{eff}} \to +\infty$ both as $u \to 0$ (because $-\ln u \to +\infty$) and as $u \to \infty$. Its minimum is at $2u - \dfrac{2\mu}{h^2 u} = 0$, i.e. $u_0 = \sqrt{\mu}/h$.
- **(i)** Escape is impossible for any finite energy. **(ii)** Reaching $r = 0$ is impossible. **(iii)** Every orbit with $E^* > W_{\text{eff}}(u_0)$ oscillates between two finite turning points; $E^* = W_{\text{eff}}(u_0)$ is the circular orbit $r = h/\sqrt{\mu}$.

**Case $\alpha = 3$: $F \propto 1/r^3$**

- $V_0 = -\dfrac{\mu}{2r^2} = -\dfrac{\mu}{2}u^2$. Binet: $u'' + \lambda u = 0$ with $\lambda \equiv 1 - \mu/h^2$. Then $W_{\text{eff}} = \lambda u^2$ and $(u')^2 + \lambda u^2 = E^*$.
- **$h^2 > \mu$ ($\lambda = \kappa^2 > 0$, $\kappa = \sqrt{1 - \mu/h^2}$):** $u = A\cos(\kappa\theta + \delta)$ with $A = \sqrt{u_0^2 + (u_0'/\kappa)^2}$. The cosine vanishes at a finite angle, so $u \to 0$ in every case: **(i) the particle always escapes**; a particle moving inward ($u_0' > 0$) is first reflected at the periapsis $u = A$. **(ii)** never, since $u \le A$. **(iii)** no oscillation between two positive values.

- $h^2 < \mu$ **($\lambda = -\kappa^2 < 0$, $\kappa = \sqrt{\mu/h^2 - 1}$):** $u = C_1 e^{\kappa\theta} + C_2 e^{-\kappa\theta}$ with $C_1 = \frac{1}{2}(u_0 + u_0'/\kappa)$, $C_2 = \frac{1}{2}(u_0 - u_0'/\kappa)$. If $C_1 > 0$ (i.e. $u_0' > -\kappa u_0$), $u \to \infty$: **(ii) the particle reaches the origin**. If $C_1 < 0$ ($u_0' < -\kappa u_0$), $u$ reaches 0 at a finite angle: **(i) escape**. If $C_1 = 0$, $u = u_0 e^{-\kappa\theta} \to 0$ only asymptotically (spiral $r = r_0 e^{\kappa\theta}$), a limiting case of escape. **(iii)** no oscillation.
- $h^2 = \mu$ **($\lambda = 0$):** $u'' = 0 \Rightarrow u = u_0 + u_0'\theta$ (reciprocal spiral). $u_0' > 0$: reaches the origin; $u_0' < 0$: escapes at $\theta = u_0/|u_0'|$; $u_0' = 0$: circle $r = 1/u_0$.

**Case $\alpha = 4$:** $F \propto 1/r^4$

- Binet: $u'' + u = \dfrac{\mu}{h^2}u^2$. $V_0 = -\dfrac{\mu}{3}u^3$, so $W_{\text{eff}} = u^2 - \dfrac{2\mu}{3h^2}u^3 = u^2\left(1 - \dfrac{2\mu}{3h^2}u\right)$.
- $W_{\text{eff}}'(u) = 2u - \dfrac{2\mu}{h^2}u^2 = 2u\left(1 - \dfrac{\mu}{h^2}u\right)$: stationary points $u = 0$ (minimum, $W = 0$) and $u_b = h^2/\mu$ (maximum, barrier $W_{\max} = u_b^2 - \dfrac{2\mu}{3h^2}u_b^3 = \dfrac{h^4}{3\mu^2}$). For $u > u_b$, $W_{\text{eff}}$ decreases without bound and crosses zero at $u_0 = 3h^2/(2\mu) > u_b$.
- **There is no interior minimum**, so the allowed set $\{W_{\text{eff}} \le E^*\}$ either contains $u = 0$ or extends to $u \to \infty$. Because $\dot\theta = hu^2 > 0$, $u'$ has the sign of $\dot u$.
- $0 \le E^* < W_{\max}$**:** two roots $u_1 \in [0, u_b)$ and $u_2 > u_b$; allowed $u \le u_1$ or $u \ge u_2$. If $u(0) \le u_1$ the particle is reflected at $u_1$ and escapes; if $u(0) \ge u_2$ it is reflected at $u_2$ and falls to the origin.
- $E^* < 0$**:** only $u \ge u_2 > u_0$ is allowed; the particle reaches $r = 0$. $E^* > W_{\max}$**:** no turning points; $u'(0) < 0$ (moving outward) escapes and $u'(0) > 0$ (moving inward) falls to the origin. $E^* = W_{\max}$**:** the unstable circular orbit $u = u_b$ and asymptotic approaches to it.
- **(i)** escape: $0 \le E^* < W_{\max}$ with $u(0) < u_b$, or $E^* > W_{\max}$ with $u'(0) < 0$. **(ii)** reaching $r = 0$: $E^* < 0$, or $0 \le E^* < W_{\max}$ with $u(0) > u_b$, or $E^* > W_{\max}$ with $u'(0) > 0$. **(iii)** bounded oscillation between two values of $u$: **does not occur**; it would need an interior minimum of $W_{\text{eff}}$, and the only bounded motion is the unstable circle $u = u_b$.

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Dimensional check:** $[F/m] = \mathrm{m/s^2} = [\mu]/\mathrm{m}^\alpha$ gives $[\mu] = \mathrm{m}^{\alpha+1}/\mathrm{s}^2$. With $[h^2] = \mathrm{m}^4/\mathrm{s}^2$ and $[u] = \mathrm{m}^{-1}$, $\dfrac{\mu}{h^2}u^{\alpha-2}$ has units $\mathrm{m}^{\alpha-3}\,\mathrm{m}^{-(\alpha-2)} = \mathrm{m}^{-1}$, the same as $u$ and $u''$ ($\theta$ is dimensionless).
- **Stability of circular orbits (consequence of the first integral):** for general $\alpha$, $W_{\text{eff}}'(u_0) = 0$ gives $u_0^{\alpha-3} = h^2/\mu$ and $W_{\text{eff}}''(u_0) = 2(3 - \alpha)$, so the circular orbit is a minimum (stable) for $\alpha < 3$ and a maximum (unstable) for $\alpha > 3$, in agreement with the cases above ($\alpha = 4$: barrier at $u_b = h^2/\mu$).

- **Numerical check** (Python, RK4 with step $10^{-4}$ rad on $u'' = -u + (\mu/h^2)u^{\alpha-2}$, units with $\mu = h = 1$ unless stated):

| Case | Initial data $(u_0, u_0')$ | Predicted | Integration result |
|---|---|---|---|
| $\alpha = 2$ | $(1.5, 0)$ | bounded, 0.5 to 1.5 | $u \in [0.5000, 1.5000]$ over 60 rad |
| $\alpha = 2$ | $(0.5, -0.9)$, $E^* > 0$ | escape | $u = 0$ at $\theta = 0.823$ |
| $\alpha = 4$ | $(0.5, 0)$, $E^* = 1/6 < 1/3$ | reflected, then escape | $u = 0$ at $\theta = 2.078$ |
| $\alpha = 4$ | $(1.2, 0)$, $E^* = 0.288 < 1/3$ | collapse | $u \to \infty$ at $\theta = 3.713$ |
| $\alpha = 4$ | $(0.5, 0.9)$, $E^* > 1/3$ | collapse | $u \to \infty$ at $\theta = 3.258$ |
| $\alpha = 4$ | $(0.5, -0.9)$, $E^* > 1/3$ | escape | $u = 0$ at $\theta = 0.524$ |
| $\alpha = 4$ | $(1, 0)$, $E^* = W_{\max}$ | unstable circular orbit | $u = 1$ for 20 rad |
| $\alpha = 3$, $\mu/h^2 = 2$ | $(1, 0.2)$, $C_1 > 0$ | collapse | $u = \cosh\theta + 0.2\sinh\theta$ is finite at every finite $\theta$ and $\to \infty$ only as $\theta \to \infty$ (the particle reaches $r = 0$ in finite time after infinitely many turns); $\theta = 4.423$ is the integrator cutoff |
| $\alpha = 3$, $\mu/h^2 = 2$ | $(1, -1.5)$, $C_1 < 0$ | escape | $u = 0$ at $\theta = 0.805$ |
| $\alpha = 3$, $\mu/h^2 = 1/2$ | $(1, 0)$ | escape | $u = 0$ at $\theta = 2.222$ |
| $\alpha = 3$, $\mu = h^2$ | $(1, -0.1)$ | escape at $\theta = 10$ | $u = 0$ at $\theta = 10.000$ |
| $\alpha = 1$ | $(0.5, 0.3)$ | bounded | $u \in [0.471, 1.653]$ over 100 rad |

The official key (Problems.pdf) gives no answer for Problem 42, so these checks are against the derived classification only.

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*Source: Problems.pdf (PDF page 29); no key available*

PROBLEM 43

# Two Connected Particles through a Hole

Original statement · Problems.pdf, p. 30

Angular Momentum + Energy

We consider a smooth horizontal plane with a small hole in it. One of the ends of a massless cord of length $2a$ is passed through the hole and then, two point particles, $P_1$ and $P_2$, each with the same mass $m$, are attached to the ends of the cord. Therefore, $P_1$ is constrained to move on the plane and $P_2$ moves along a vertical line below the hole.

Initially, half of the cord lies under the plane and half of the cord is on the plane (so $r(0) = a$). At that instant, $P_1$ is pushed with a speed $v_0$ perpendicular to the cord, counterclockwise as seen from above. Assume that the cord is always in tension.

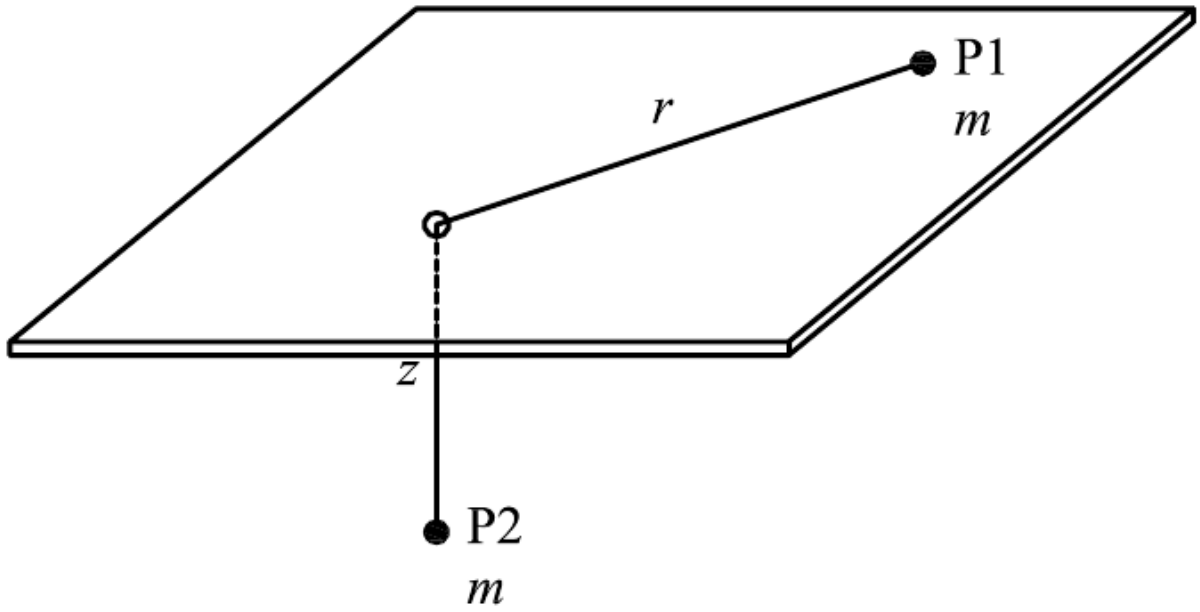



*Figure: Smooth horizontal plane with a small hole, particle P1 on the plane at distance r from the hole and particle P2 hanging below it, both of mass m (Problems.pdf, p. 30).*

**(a)** What is the value of $v_0$ if $P_1$ describes a circumference?

**(b)** In that case, what is the value of the tension in the cord?

**(c)** If $v_0 = \sqrt{8ag/3}$, reduce the problem to integrals (i.e., quadratures).

**(d)** Show that for this value of $v_0$, $P_1$ oscillates between $r = a$ and $r = 2a$.

**(e)** Apply Newton's law to $P_2$ and show that the tension of the cord is $T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$.

**(f)** Write down the new equations of motion if the plane on which $P_1$ moves is rough, with a friction coefficient $\mu$.

*Hint: the roots of the polynomial $-3x^3 + 7x^2 - 4$ are $x_1 = 1$, $x_2 = 2$ and $x_3 = -2/3$.*

**Worked Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **System:** $P_1$ on a smooth horizontal plane at polar position $(r, \theta)$ and $P_2$ hanging on the vertical through the hole $O$, joined by an inextensible massless cord of length $2a$ passing through $O$.
- **Constraints and degrees of freedom:** the cord length gives $r + \ell_2 = 2a$, where $\ell_2$ is the hanging length, so $z_2 = -\ell_2 = r - 2a$ ($Oz$ up). The generalized coordinates are $(r, \theta)$, $\mathrm{CDOF} = 2$.
- **Hypotheses:** inertial frame $S_0$ with $O$ at the hole; smooth hole and plane for parts (a)-(e); the cord stays taut; no drag.
- **Initial state:** $r(0) = a$, $\dot{r}(0) = 0$, $r(0)\dot{\theta}(0) = v_0$, hence $\dot{\theta}(0) = v_0/a$ and $\dot{z}_2(0) = 0$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $m$ | mass of each particle | kg |
| $a$ | half of the cord length | m |
| $v_0$ | initial speed of $P_1$ | m/s |
| $g$ | gravitational acceleration | m/s$^2$ |

| Symbol | Meaning | SI unit |
|---|---|---|
| $\mu$ | friction coefficient (part f) | 1 |

PHASE 2

### Reference Frames, Vector Bases & Coordinate Geometry

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$, horizontal plane $z = 0$, $\mathbf{k}$ up.
- **Change of basis (polar basis of $P_1$).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

with $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}$, $\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot\theta\,\mathbf{e}_r$ (Poisson).

- **Kinematics:** $\mathbf{r}_1 = r\mathbf{e}_r$, $\mathbf{v}_1 = \dot r\mathbf{e}_r + r\dot\theta\mathbf{e}_\theta$; $\mathbf{r}_2 = (r-2a)\mathbf{k}$, $\mathbf{v}_2 = \dot r\mathbf{k}$.

$$T_{\text{kin}} = \tfrac{1}{2}m\left(\dot r^2 + r^2\dot\theta^2\right) + \tfrac{1}{2}m\dot r^2 = m\dot r^2 + \tfrac{1}{2}mr^2\dot\theta^2, \qquad V = mgz_2 = mg(r-2a)$$

PHASE 3

### Step-by-Step Mathematical Deduction

**(a) Circular motion, $r \equiv a$**

> **Pedagogical Justification:** On the plane the only horizontal force on $P_1$ is the tension $-T\mathbf{e}_r$, which is central. For $r = a$ the radial acceleration is purely centripetal; $P_2$ must stay at rest.

$$P_1:\ m\left(\ddot r - r\dot\theta^2\right) = -T \xrightarrow{r=a,\ \dot r=0} T = ma\dot\theta^2 = \frac{mv_0^2}{a}, \qquad P_2:\ m\ddot z_2 = T - mg = 0 \implies T = mg$$

$$\frac{mv_0^2}{a} = mg \implies v_0 = \sqrt{ag}$$

**(b) Tension in the circular case**

$$T = mg$$

**(c) Quadratures for $v_0 = \sqrt{8ag/3}$**

> **Pedagogical Justification:** The tension on $P_1$ is central, so $h = r^2\dot\theta$ is constant; the plane and hole are smooth and the cord inextensible, so $E = T_{\text{kin}} + V$ is constant. The two first integrals leave one equation for $r(t)$.

$$h = r^2\dot\theta = av_0 = a\sqrt{\frac{8ag}{3}} \implies h^2 = \frac{8}{3}a^3 g$$

$$E = m\dot r^2 + \frac{mh^2}{2r^2} + mg(r - 2a), \qquad E(0) = \tfrac{1}{2}m\cdot\frac{8ag}{3} - mga = \frac{1}{3}mga$$

Equating, dividing by $m$ and isolating $\dot r^2$:

$$\dot r^2 + \frac{4a^3 g}{3r^2} + gr - 2ga = \frac{1}{3}ga \implies \dot r^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2} = \frac{g}{3r^2}\left(7ar^2 - 3r^3 - 4a^3\right)$$

Separating variables and applying Barrow's rule from $r(0) = a$:

$$\frac{dr}{dt} = \pm\sqrt{\frac{g}{3}}\,\frac{\sqrt{-3r^3 + 7ar^2 - 4a^3}}{r} \implies \int_a^r \frac{r'\,dr'}{\sqrt{-3r'^3 + 7ar'^2 - 4a^3}} = \pm\sqrt{\frac{g}{3}}\,t$$

**(d) Oscillation between $r = a$ and $r = 2a$**

> **Pedagogical Justification:** Motion requires $\dot{r}^2 \geq 0$, i.e. $P(r) = -3r^3 + 7ar^2 - 4a^3 \geq 0$. With $x = r/a$ the polynomial is $a^3(-3x^3 + 7x^2 - 4)$, and the hint gives its roots.

$$-3x^3 + 7x^2 - 4 = -(x-1)(x-2)(3x+2)$$

Check of the roots: $x = 1$: $-3 + 7 - 4 = 0$; $x = 2$: $-24 + 28 - 4 = 0$; $x = -2/3$: $\frac{8}{9} + \frac{28}{9} - 4 = 0$. Since $r > 0$, $3x + 2 > 0$ and $P \geq 0$ requires $(x-1)(x-2) \leq 0$:

$$1 \leq x \leq 2 \iff a \leq r(t) \leq 2a$$

The turning points ($\dot{r} = 0$) are $r_{\min} = a$ and $r_{\max} = 2a$, and the motion oscillates between them.

**(e) Tension of the cord**

> **Pedagogical Justification:** Apply Newton's second law to $P_2$ (weight and tension) with $\ddot{z}_2 = \ddot{r}$, because $z_2 = r - 2a$.

$$m\ddot{r} = T - mg \implies T = m\left(g + \ddot{r}\right)$$

Differentiate $\dot{r}^2 = \frac{7}{3}ga - gr - \frac{4a^3 g}{3r^2}$ with respect to time (chain rule, $\frac{d}{dt} r^{-2} = -2r^{-3}\dot{r}$):

$$2\dot{r}\ddot{r} = \left(-g + \frac{8a^3 g}{3r^3}\right)\dot{r} \implies \ddot{r} = -\frac{g}{2} + \frac{4a^3 g}{3r^3}$$

$$\boxed{T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)}$$

$T > 0$ for all $r \in [a, 2a]$, so the cord stays taut.

**(f) Rough plane**

> **Pedagogical Justification:** Coulomb friction acts on $P_1$ opposite to its velocity, with magnitude $\mu N_1$. The normal force is $N_1 = mg$, because the cord is horizontal on the plane and only the weight acts vertically.

$$\mathbf{F}_f = -\mu m g \, \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = -\mu m g \, \frac{\dot r \, \mathbf{e}_r + r\dot\theta \, \mathbf{e}_\theta}{\sqrt{\dot r^2 + r^2\dot\theta^2}}$$

$$\mathbf{e}_r : \; m\left(\ddot r - r\dot\theta^2\right) = -T - \mu m g \frac{\dot r}{\sqrt{\dot r^2 + r^2\dot\theta^2}}$$

$$\mathbf{e}_\theta : \; m\left(r\ddot\theta + 2\dot r\dot\theta\right) = -\mu m g \frac{r\dot\theta}{\sqrt{\dot r^2 + r^2\dot\theta^2}}, \qquad P_2 : \; m\ddot r = T - mg$$

Adding the $\mathbf{e}_r$ equation and the $P_2$ equation eliminates $T$; multiplying the $\mathbf{e}_\theta$ equation by $r$ gives the evolution of $h$:

$$2\ddot r - r\dot\theta^2 = -g - \mu g \frac{\dot r}{\sqrt{\dot r^2 + r^2\dot\theta^2}}, \qquad \frac{d}{dt}\left(r^2\dot\theta\right) = -\mu g \, \frac{r^2\dot\theta}{\sqrt{\dot r^2 + r^2\dot\theta^2}}$$

Angular momentum and energy are no longer conserved (friction dissipates energy), so the problem cannot be reduced to the quadratures of (c).

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Tension at the turning points:** $T(a) = mg\left(\frac{1}{2} + \frac{4}{3}\right) = \frac{11}{6}mg$ and $T(2a) = mg\left(\frac{1}{2} + \frac{4}{24}\right) = \frac{2}{3}mg$, both positive.

- **Consistency with (a):** for $v_0^2 = ag$, $h^2 = a^3 g$ and $\ddot{r}(a) = -\frac{g}{2} + \frac{h^2}{2a^3} = 0$, so $r = a$ is an equilibrium of the radial equation, as expected for the circular case.
- **Radial period** (consequence of (c) and (d)): with $x = r/a$ and $x = \frac{3}{2} - \frac{1}{2}\cos s$, $(x-1)(2-x) = \frac{1}{4}\sin^2 s$ and $dx = \frac{1}{2}\sin s\, ds$, so $\tau_r = 2\sqrt{\dfrac{3a}{g}} \displaystyle\int_0^{\pi} \frac{x(s)\, ds}{\sqrt{3x(s) + 2}} \approx 6.342\sqrt{\frac{a}{g}}$.
- **Numerical check** (Python, RK4 on $\ddot{r} = -g/2 + 4a^3 g/(3r^3)$ with $a = g = 1$, $r(0) = 1$, $\dot{r}(0) = 0$): maximum radius $2.0000$, period $6.3425$ s (the quadrature gives $6.3424$), and the energy-integral residual stays below $4 \times 10^{-14}$.
- **Dimensions:** $[T] = [m][g] = \mathrm{N}$ and $[v_0] = \sqrt{[a][g]} = \mathrm{m/s}$.
- **Key:** the official key (Problems.pdf) has no entry for Problem 43.

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*Source: Problems.pdf (PDF page 30); no key available*

PROBLEM 44

# The Little Prince Throwing Seeds

Original statement · Problems.pdf, p. 31

Angular Momentum + Vis-Viva

The little Prince is on his spherical asteroid, which has radius $R$ and mass $M$, and does not rotate. While he sits at the equator, he throws a baobab seed with an initial velocity of magnitude $v_0$ and angle $\beta = 60^\circ$ with respect to the local horizontal. Knowing that it reaches an apocenter of $r_a = 3R$ before falling back, determine $v_0$ as a function of the parameters of the problem and the universal gravitational constant $G$.

## Worked Solution (4-Phase Methodology)

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **System:** a seed of mass $m \ll M$ launched from the surface ($r_0 = R$) with speed $v_0$ at $60^\circ$ above the local horizontal; it reaches the apocenter $r_a = 3R$.
- **Hypotheses:** spherically symmetric, non-rotating asteroid, so the gravitational force $\mathbf{F} = -\dfrac{GMm}{r^2}\mathbf{e}_r$ is central and conservative with $\mu = GM$; vacuum (no drag); inertial frame $S_0$ centred on the asteroid.
- **Degrees of freedom:** free particle, planar motion by central-force conservation of $\mathbf{H}_O$; coordinates $(r, \theta)$.
- **Initial state:** $r = R$, radial component $v_{r0} = v_0 \sin 60^\circ = \frac{\sqrt{3}}{2} v_0$ and transverse component $v_{\theta 0} = v_0 \cos 60^\circ = \frac{1}{2} v_0$.

| Symbol | Meaning | SI unit |
|---|---|---|
| $M, R$ | mass and radius of the asteroid | kg, m |
| $G$ | universal gravitational constant, $6.6743 \times 10^{-11}$ | $\mathrm{m^3/(kg\ s^2)}$ |
| $\mu = GM$ | gravitational parameter | $\mathrm{m^3/s^2}$ |
| $\beta$ | launch angle above the horizontal | $60^\circ$ |

| Symbol | Meaning | SI unit |
|---|---|---|
| $r_a$ | apocenter radius | $3R$ |

PHASE 2

### Reference Frames, Vector Bases & Coordinate Geometry

- **Change of basis (polar basis in the orbital plane).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis rotated by the true anomaly $\theta$ about $\mathbf{k}$ (along $\mathbf{H}_O$):

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1, \qquad [{}_0R_1]^T[{}_0R_1] = I$$

- **Velocity in $\mathcal{B}_1$:** $\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta$; at launch $\dot r = v_0\sin 60^\circ$ and $r\dot\theta = v_0\cos 60^\circ$. At the apocenter $\dot r = 0$, so the velocity is purely transverse.

PHASE 3

### Step-by-Step Mathematical Deduction

#### 1. Conservation of specific angular momentum

**Pedagogical Justification:** The gravitational force passes through $O$, so $\mathbf{M}_O = \mathbf{0}$ and $h = \|\mathbf{r}\times\mathbf{v}\| = r^2\dot\theta$ is constant. Evaluate it at launch and at the apocenter, where $v_a = r_a\dot\theta_a$.

$$h = R\,(v_0\cos 60^\circ) = \frac{1}{2}Rv_0, \qquad h = r_a v_a = 3Rv_a \implies v_a = \frac{v_0}{6}$$

**2. Conservation of specific energy**

> **Pedagogical Justification:** Gravity is conservative with $V_0 = -\mu/r$ (derived with Barrow's rule, reference at infinity), so $\xi = \frac{1}{2}v^2 - GM/r$ takes the same value at launch and at the apocenter.

$$\frac{1}{2}v_0^2 - \frac{GM}{R} = \frac{1}{2}\left(\frac{v_0}{6}\right)^2 - \frac{GM}{3R} = \frac{v_0^2}{72} - \frac{GM}{3R}$$

**3. Solving for $v_0$**

$$\left(\frac{1}{2} - \frac{1}{72}\right)v_0^2 = \left(1 - \frac{1}{3}\right)\frac{GM}{R} \implies \frac{35}{72}v_0^2 = \frac{2}{3}\frac{GM}{R} \implies v_0^2 = \frac{2}{3}\cdot\frac{72}{35}\frac{GM}{R} = \frac{48}{35}\frac{GM}{R}$$

$$\boxed{v_0 = \sqrt{\frac{48}{35}\frac{GM}{R}}}$$

PHASE 4

## Physical Interpretation & Dimensional Verification

- **Bound orbit:** the escape speed at the surface is $v_e = \sqrt{2GM/R}$, i.e. $v_e^2 = \frac{70}{35}GM/R$. Since $\frac{48}{35} < 2$, $v_0 < v_e$ and the orbit is an ellipse: the seed falls back.
- **Orbital parameters:** $h^2 = \frac{1}{4}R^2v_0^2 = \frac{12}{35}GMR$, $p = h^2/\mu = \frac{12}{35}R$. From $r_a = p/(1-e) = 3R$: $1 - e = \frac{4}{35}$ and $e = \frac{31}{35} \approx 0.8857$. The periapsis of the full conic is $r_p = p/(1+e) = \frac{2}{11}R < R$, inside the asteroid; only the arc from the launch point to the apocenter is physical, and the seed lands before reaching the periapsis.
- **Independent check with the Binet solution.** With $u = \frac{\mu}{h^2}(1 + e\cos\theta)$ and the launch values $u(\theta_0) = 1/R$, $u'(\theta_0) = -\dot{r}/h$ (here $\dot{r} = \frac{\sqrt{3}}{2}v_0$), the eccentricity formula of the theory page gives

$$e = \frac{h^2}{\mu}\sqrt{\left(\frac{1}{R} - \frac{\mu}{h^2}\right)^2 + \left(\frac{\dot r}{h}\right)^2} = \frac{12R}{35}\sqrt{\frac{529}{144R^2} + \frac{3}{R^2}} = \frac{12R}{35}\cdot\frac{31}{12R} = \frac{31}{35}$$

and the launch angle follows from $\tan\beta = \dot r/(r\dot\theta) = \dfrac{R}{p}\,e\sin\theta_0$ with $\cos\theta_0 = \dfrac{p/R - 1}{e} = -\dfrac{23}{31}$, $\sin\theta_0 = \dfrac{\sqrt{432}}{31}$:

$$\tan\beta = \frac{35}{12}\cdot\frac{31}{35}\cdot\frac{\sqrt{432}}{31} = \frac{\sqrt{432}}{12} = \sqrt{3} \implies \beta = 60^\circ$$

- **Numerical check** (Python, velocity-Verlet integration of $\ddot{\mathbf r} = -\mu\mathbf r/r^3$ with $\mu = R = 1$ and $v_0 = \sqrt{48/35}$): the maximum radius reached is $2.99999999981R$, in agreement with $r_a = 3R$.
- **Dimensions:** $[v_0] = \sqrt{[G][M]/[R]} = \sqrt{\dfrac{\mathrm{m}^3}{\mathrm{kg\,s}^2}\cdot\dfrac{\mathrm{kg}}{\mathrm{m}}} = \mathrm{m/s}$.
- **Key:** the official key (Problems.pdf) has no entry for Problem 44.

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*Source: Problems.pdf (PDF page 31); no key available*